École Polytechnique Fédérale de Lausanne

Physical Decoupled Multimodal Vision-Bassed Programmable Skin

Master thesis

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Contents

1 Introduction **1**

2 Experimental Setup **3**
2.1 Configuration of the Different Physical Components 3
2.2 Elastomer 4
2.3 Grating 4
2.3.1 Glue 5
2.4 Thermochromic Liquid Crystals Powder 5
2.4.1 Sensitive Range (25°C-31°C) 5
2.4.2 Wide Range (10°C-70°C) 6

3 Methods **6**
3.1 Fourier Transform Profilometry 6
3.2 Temperature-based Segmentation 8

4 Results **9**
4.1 Temperature: Color Stripes 10
4.1.1 Experiment Setup 10
4.1.2 Results 10
4.2 Temperature: Black Stripes 10
4.2.1 Experiment Setup 10
4.2.2 Results 10
4.3 Temperature: Model 11
4.4 Force-Depth Relationship 13
4.4.1 Experiment Setup 13
4.4.2 Results 13
4.5 Stripe Width Ratio-Force relationship 14
4.5.1 Experiment Setup 14
4.5.2 Analysis & Results 15
4.6 3D Reconstruction 15
4.7 Integrated Pipeline 16

5 Potential Applications **18**

6 Conclusion **20**

7 References **21**

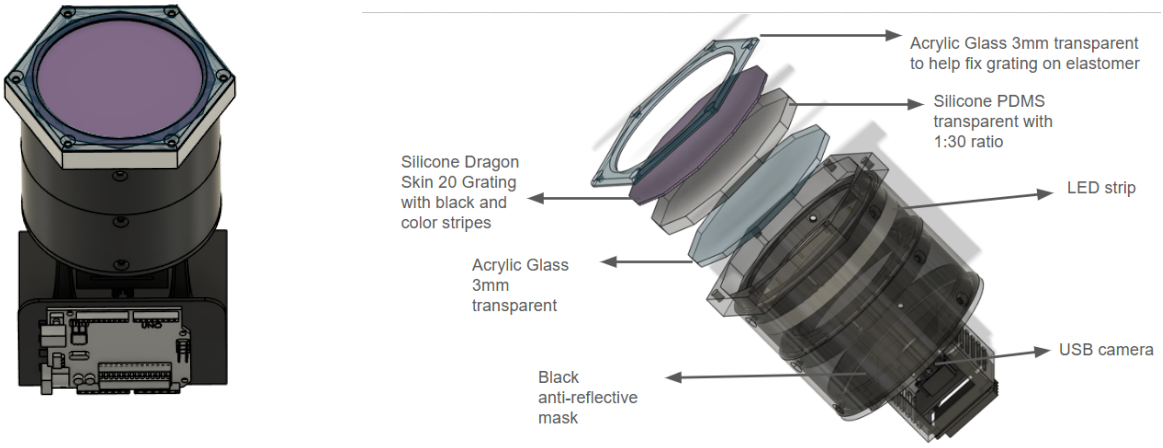
A Appendix **22**
A.1 Brightness of the LED strip 22
A.2 Colors observed in the Temperature Experiments 22

A.3 Masses used for fabrication 24
A.4 Code 24

Abstract

Inspired by the feeling of touch that is present across humans, this project presents a soft multi-modal decoupled vision based tactile and temperature sensor. A short and sensitive range of temperatures that stand between room and body temperature (25°C-31°C), as well as a broader range of temperatures (10°C-70°C), can be measured by combining thermochromic liquid crystal powders (pigments) into a thin layer of silicone. A periodic stripe pattern allows for the use of Fourier Transform Profilometry (FTP), helping to establish shape, force and depth which can then be visualized in 3D. As these separate concepts work without depending on the other, it is considered to be decoupled which allows the sensor to measure either one of the aspects or both when provided with a reference (non-deformed) image and a deformed image (where the sensor is impacted by physical or temperature change).

1 Introduction



(a) Layout of the exterior

(b) Annotation of the different components on the inside that make up the core of the sensor

Figure 1: Design overview showing all components

Touch is one of the five senses that human beings have, which are vital for us to be able to understand our environment. Trying to get robots to replicate the many sensations that touch can bring (which allow humans to feel shape, pressure, force, temperature, among other things) would greatly help with their manipulation and dexterity, while improving safety in interactions with humans.[1]

The world of tactile sensors is a field where research has greatly increased over the past decade or so.[1] Many of these sensors have been developed with electronic sensors such as piezoresistive or capacitive sensors, which add much complexity to the manufacturing process and in which a single component having a problem will cause the entire sensor to have issues. [2, 3]. An alternative to this is for tactile sensors to be based on vision. Vision-based tactile sensors are hugely adaptable, they have a high resolution, and also have a straightforward wiring and manufacturing process. [4] They can come in a variety of forms, but the ideal make-up consists of four different components: a contact module, an illumination module, a perception module, and a base module. The contact module serves to interact with the environment and is usually soft, the illumination model serves to provide the correct lighting for the camera, the perception

module is the camera and associated features, while the base module are the components which give the sensor structure and tie the first three modules together.[5]

In the world of vision-based tactile sensors, GelSight[6, 7] and TacTip[8] are among those that dominate and have led the field in recent times.[9] The goal of GelSight is to be able to use shape and texture of a surface to then make a 3D reconstruction of it.[7] In order to do this, they popularized the following hardware format: a clear elastomer that lays atop acrylic glass, uniform LED lights to provide illumination, and a camera behind the acrylic glass to be able to capture pictures of the deformation.[10] Their elastomer includes black spots as markers which can detect contact force and torque. The motion even allows one to view different types of force (such as normal and shear) or even torque (in-plane or tilted). [7]

Since then, there have been a wave of vision-based sensors that have focused on contact force. Some, however, have also looked at other aspects such as temperature, like HaptiTemp [11] or in the vision-based soft somatosensory system of Yu et al. (2020) which tries to incorporate the detection of both those features.[12] HaptiTemp has a similar physical structure to GelSight, but here its goal is to determine the potential temperature range based on thermochromic pigments. They go from a given color to colorless when they hit a certain activation temperature, and so thermochromic pigments of different colors that have separate activation temperatures can then be mixed together in order to detect different ranges of temperature.[11] The vision-based soft somatosensory system tries to be able to measure both contact force and temperature, by using a white and black grating for the contact force and thermochromic pigments to measure temperature. [12]

This project seeks to build on those sensors by introducing a physical decoupled programmable skin. It is also a vision-based tactile sensor, but it is multimodal by nature with a single camera and no electrodes. The sensor is split between two main functions: temperature and tactile. In the temperature domain, it is able to estimate the temperature across the region of interest, and segment any shapes caused by the transfer of heat of an object. For the tactile branch, the sensor is able to measure the contact force, depth of contact, shape, and a 3D reconstruction can then be made.

The temperature component utilizes thermochromic pigments at two different ranges (25°C-31°C) and (10°C-70°C) which is possible due to being mixed into the grating, which is the first of our novel contributions. This demonstrates how the sensor can act more precisely within a small range that is close to body temperature, while it is also possible for this sensor to measure a wider temperature range in order to increase its flexibility and potential uses, such as being able to improve its recognition accuracy due to the thermal conductivity differing across materials.[13]

The tactile element of the sensor uses Fourier Transform Profilometry (FTP) [14] at its core. It is through the use of phase differences in FTP that the sensor is able to determine the shape and the force being applied by the contact object. It is then by using the relationship between force and depth that we are able to have a 3D reconstruction, which is another novel contribution of this project. Finally, the biggest breakthrough for this project is being able to combine the two aspects and algorithms in a decoupled manner, which allows the sensor to detect both temperature induced changes and force based changes in a single image when compared with a reference image.

2 Experimental Setup

Figure 2: The setup of the sensor

2.1 Configuration of the Different Physical Components

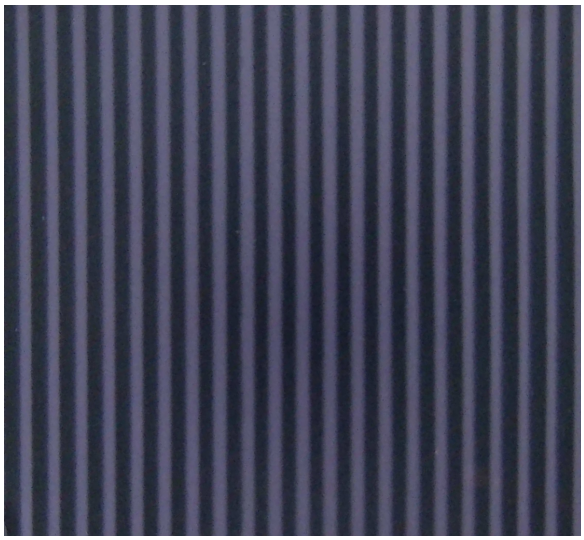
The setup can be seen in Figure 2. The perception module in this setup is a fixed-focus USB camera and is placed to face the grating. The base of the setup is covered by a black 3D printed disk with a hole in the middle, where the inner diameter is set to be slightly bigger than the lens of the camera. This is to prevent reflection of the base on the acrylic glass. The illumination module is a white light LED strip that follows the circular interior of the setup where the LEDs face inward (so the strip is vertical). The brightness of the LED strip is set by an Arduino where PWM is used in order for the light to not be emitted at full brightness. This helps improve the contrast between the colored and black stripes which is needed for the tactile component. The contact module has an elastomer that is placed atop the acrylic glass, with the grating being then glued to the elastomer. Finally, the base module is entirely 3D printed in black, except for the top part which is white. This was to help give a clearer image, without there being too many reflections being seen with the camera due to the acrylic glass and the overall glare from the white core.

2.2 Elastomer

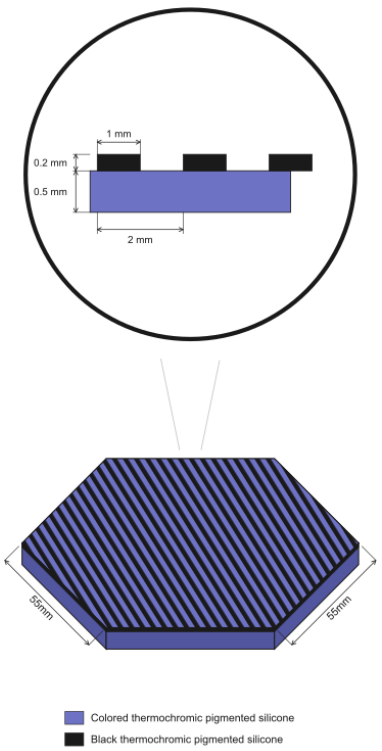
The elastomer was made with a variant of silicone called polydimethylsiloxane (PDMS). The PDMS is mixed with a curing agent that allows it to solidify, and the ratio of PDMS to curing agent defines how hard or soft the elastomer is once it has cured. The ratio used in this project was 1:30 (by mass) for curing agent to PDMS respectively. This resulted in the elastomer being soft enough, which allowed the sensor to detect subtler contacts. Lower ratios result in the elastomer being very hard and rigid, which would not allow small changes in force to be measured and therefore negatively impact the overall sensitivity of the sensor.

2.3 Grating

The grating was made in two parts: the colored base and the black grating superimposed on top. The colored base was made with DragonSkin20 (Smooth On), which was used as it could be fabricated with a small thickness and showed a good amount of elasticity while being able to return to its previously undeformed state rapidly even after multiple indents were made. On the other hand, the black stripes were made with Psycho Paint (Smooth On), which proved to be the best for achieving the thin (in width and thickness) stripes required. The colored part was made with a thickness of 0.5mm (by using an adjustable film applicator). The black grating was then put on by having a 3D printed mask (0.2mm thickness and a period of 2mm) stuck onto the colored base (once cured) and then the black stripes were pressed into the mask and left to cure. The mask could then be removed once everything had cured. This was done in order to have as thin a layer as possible to have a quick transfer of heat through the grating (as it is the stripes on the underside which are visible to the camera), while still being reasonable to manufacture.



(a) View of the grating



(b) Illustration of the dimensions and thicknesses of the grating

Figure 3

In Figure 3a, an image is shown to demonstrate what the grating looks like once manufactured. It faces down and it is the striped side which faces the elastomer and camera. Figure 3b shows the dimension of the hexagonal shape that the grating takes, as well as the thickness and period of the stripes and base.

Figure 4 shows the variation of the colors with temperature, so that the visual aspect becomes more intuitive. In the first row, the change in color due to the thermochromic pigments of the colored base is shown as the temperature increases, while the second row demonstrates the same except for the black stripes. There is a clear visual difference as the temperature increases, which is what this project exploits in order to predict temperature.

2.3.1 Glue

Once the elastomer and the grating are both fully cured, the grating was glued onto the elastomer using PDMS as glue with the ratio of PDMS to curing agent being the same as when making the elastomer.

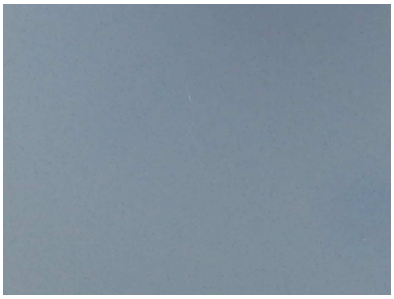
2.4 Thermochromic Liquid Crystals Powder

2.4.1 Sensitive Range (25°C-31°C)

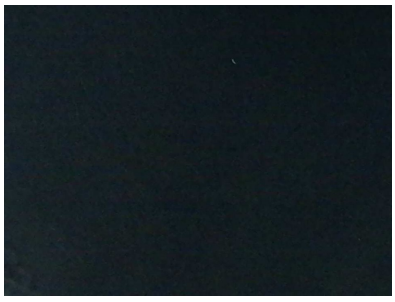
The sensitive range was created by mixing three different colors (yellow, red, and blue) of thermochromic pigments that each have a separate activation temperature (25°C, 28°C, 31°C), respectively, into the colored base of the grating (DragonSkin20). They were added with a ratio

(a) Colored DragonSkin20 at 25°C

(b) Colored DragonSkin20 at 28°C



(c) Colored DragonSkin20 at 31°C



(d) Black Psycho Paint at 25°C

(e) Black Psycho Paint at 45°C

(f) Black Psycho Paint at 70°C

Figure 4: The six pictures above are pictures of the silicone with thermochromic pigments at the ratios specified in Section 2.4 to show how the color is different at different activation temperatures for both the sensitive range (first row) and the wide range (second row)

of 5:3:3 (yellow:red:blue) by mass as this helped to give the best blend allowing for a noticeable change in color between those temperatures. White dye was also added to the mix so that once the temperature was above all the activation temperatures, the silicone would be of a white color rather than transparent which helps with the contrast in the grating with the black stripes.

2.4.2 Wide Range (10°C-70°C)

In contrast, the wide range was made by mixing a series of black thermochromic pigments with eight different activation temperatures (10°C, 18°C, 25°C, 28°C, 31°C, 38°C, 45°C, 70°C) into the black stripes of the grating (Psycho Paint). The ratio, in mass, was 5:5:6:6:6:5:5:5 respectively. This was done in order to be able to provide clear contrast with the colored stripes for the grating, yet still have a change in the darkness of the black stripes as temperature changes which would expand the capabilities of the sensor.

3 Methods

3.1 Fourier Transform Profilometry

The sensor uses FTP in order to be able to detect changes in phase to the grating and hence the subsequent shape caused by the deformation of an object. The sinusoidal fringe pattern equation can be modeled as: [14]

$$I(x,y) = A(x,y) + B(x,y)cos(2\pi f_0 x + \phi(x,y)) \tag{1}$$

where $I(x,y)$ is the intensity of the fringe image, $A(x,y)$ is the background intensity, $B(x,y)$ is the amplitude of the fringes, f_0 is the frequency of the fringe pattern and ϕ is the phase to be retrieved.[15]

In having a reference image, one can use its grayscale image to locate the fringe carrier. This is done by looking at the image in the Fourier domain where the carrier frequency, $\mathbf{f}_0 = (f_x, f_y)$, is taken as the magnitude peak and has its sign fixed by being folded into a consistent half-plane. Demodulation then occurs, which first isolates the +1 carrier lobe, then inverts the transform, and removes the carrier ramp:

$$z(\mathbf{r}) = \Big[\mathcal{F}^{-1}\{F \cdot H\}\Big] \exp\!\left(-\frac{2\pi i}{n}\,\mathbf{f}_0 \cdot \mathbf{r}\right) = A(\mathbf{r})\, e^{i\varphi(\mathbf{r})} \tag{2}$$

The equation above then gives the analytic fringe field with amplitude A and demodulated phase φ. The shape then just becomes the spatially unwrapped phase difference between the two analytic fields:

$$\Delta\varphi(\mathbf{r}) = \mathrm{unwrap}\big[\,\arg\big(z_{\mathrm{def}}\,\overline{z_{\mathrm{ref}}}\big)\big] \tag{3}$$

In order to then keep the operation modulo-2π correct prior to unwrapping, the argument of the complex conjugate product is taken. Following that, some postprocessing is done to attempt to make it less noisy and to fill any gaps that may occur.

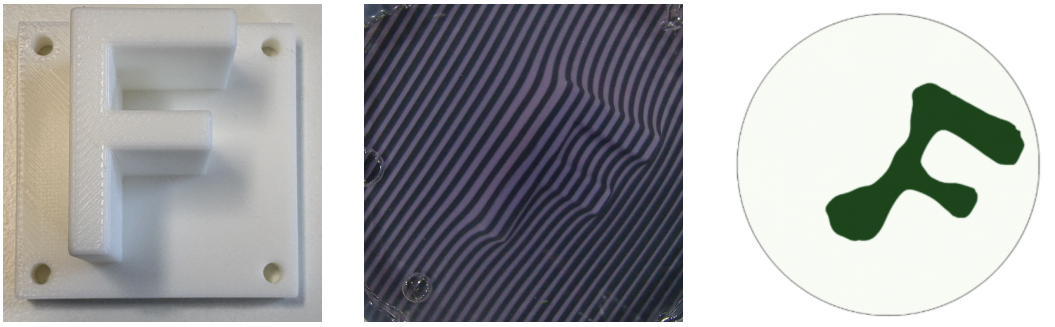
Figure 5 shows some of the shapes that were obtained using this method and how it is possible to obtain a wide variety of different shapes with it. The shapes show how the model is able to detect objects with holes in them such as a ring, which helps to increase the number of objects that can be detected such as nuts and washers. More complex shapes that were detected such as a star or the letter F also show that the sensor's detection capabilities can be expanded to

more unconventional items. The results also show that the sensor can detect separate shapes, which is a huge advantage in being able to understand what it might be interacting with.

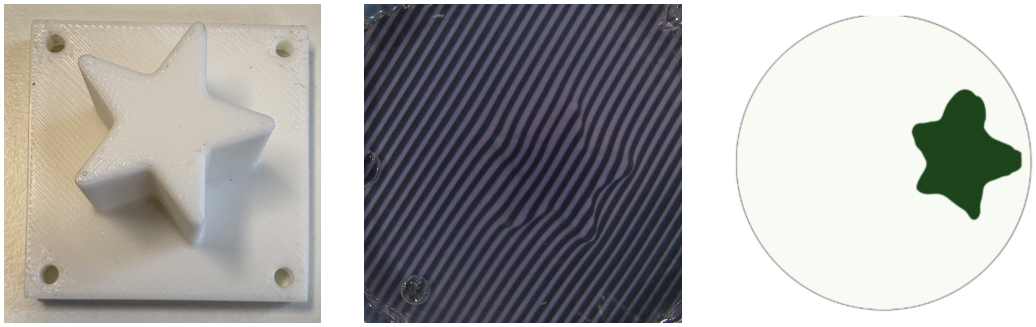
In order to better quantify how well the algorithm works in being able to capture the shapes and their locations, the IoU, Dice, Precision, and Recall scores were all calculated and can be seen in Table 1. It shows reasonably high Dice and IoU scores for all shapes, with just the three circles being a bit lower, which is probably due to the fact that they are much smaller and sharper making it more complicated for the algorithm to accurately determine its edges. The recall is overall very high however, which demonstrates the algorithm can detect these shapes very well, but overpredicts the size which leads to lower precision scores.

Shape	IoU	Dice	Precision	Recall
Square	0.7478	0.8557	0.7930	0.9292
F shape	0.711	0.831	0.919	0.758
Star	0.7273	0.8421	0.7383	0.9800
Ring	0.6919	0.8179	0.7162	0.9533
Three Circles	0.4703	0.6398	0.4718	0.9934

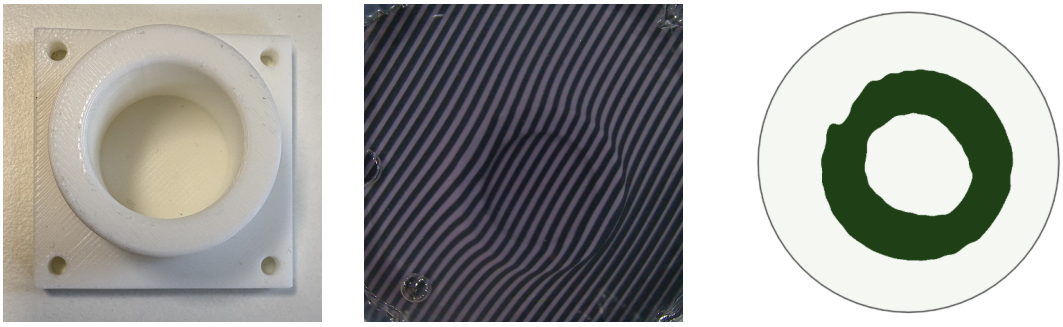
Table 1: Metrics to quantify how well the shape captures the actual shape pressed into the grating



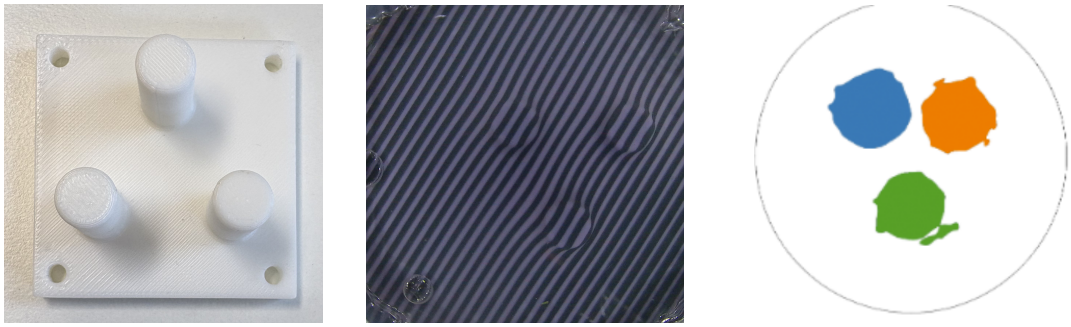
(a) F shape



(b) Star



(c) Ring



(d) Three circles

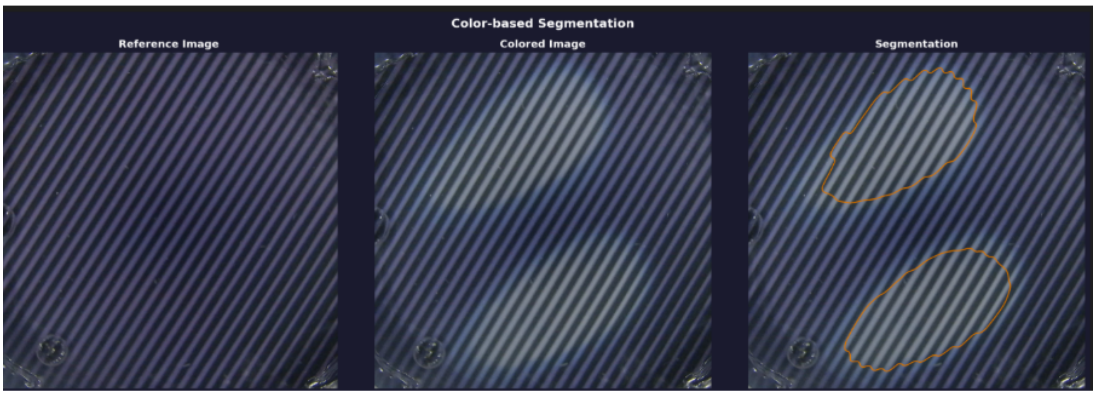
Figure 5: Shapes that have been obtained with FTP. The first column is the actual shape that was used for the indent, then the second shows a picture of the shape pressed into the grating, and the third gives the final result

3.2 Temperature-based Segmentation

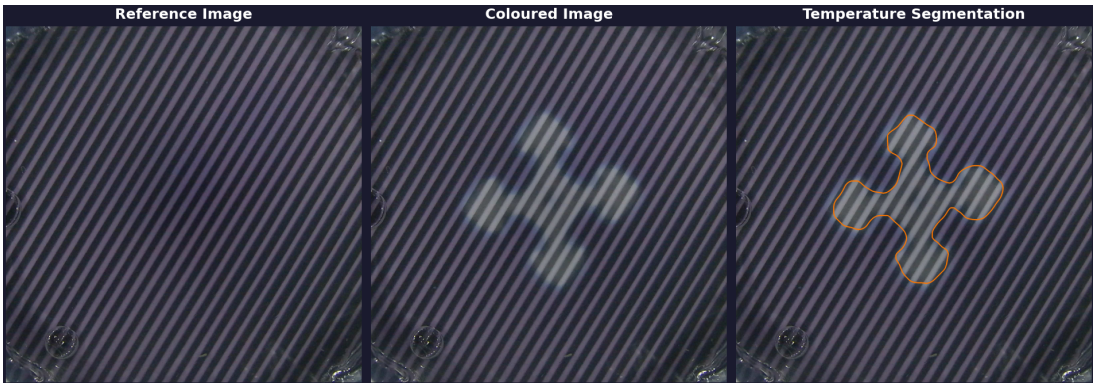
Due to the fact that the color of the grating changes as heat is applied to the sensor, we can leverage this to perform segmentation on objects. In Figure 6a, two fingerprints are shown to be

segmented when compared with a reference image. Below that, Figure 6b demonstrates a more distinct object, with clearer boundaries that it then manages to segment.

The algorithm used for the segmentation requires the image with the heated object (colored image), and an image before said object was applied to the sensor (reference image). It begins with comparing the grayscale region of interest (ROI) of the two through the ratio of the heated image to the reference image. The ratios are then normalized and the pixels with a ratio above a certain threshold that is defined by the user are considered to have been heated. This then results in the segmentation that can be seen in Figure 6.



(a)



(b)

Figure 6: Segmentation obtained based on a reference image, and an image with heat applied to the sensor

4 Results

In order for the sensor to be able to work in a decoupled manner, different experiments needed to be performed on the separate components. Tests were carried out on the sensitive range and the wide range of the thermochromic pigments to be able to then model a relationship between temperature and color. There were also tests performed to look at the relationship between force and depth, as well as stripe width and force which helps to understand how much the stripe width changes as different forces are applied to it, and consequently how that affects depth which is useful for doing a 3D reconstruction.

4.1 Temperature: Color Stripes

4.1.1 Experiment Setup

A sample that had the same properties as in the grating and the same thickness (Section 2.3, Section 2.4.1) which was glued directly to acrylic glass was used for this experiment. It was heated at different temperatures while heating and then cooling with 10 pictures taken per temperature (1 image per second). The range of temperatures taken was 24°C-70°C, where the step was of 1°C between 24°C and 32°C and of 5°C between 35°C and 70°C. The color found in each of the 10 frames at each temperature was averaged out, and the L value from the CIELAB color space was taken in order to demonstrate the change in pixel values across temperature.

4.1.2 Results

As is visible in Figure 7a, there is a clear trend in the increase in L value as the temperature increases for both the heating and cooling runs. It then flatlines above 31°C which is consistent with the fact that all activation temperatures have been reached and then it should remain white for all temperatures thereafter. Within the sensitive range, there is a hysteresis between the heating and cooling run where at equivalent temperatures the cooling run displays a much higher L value.

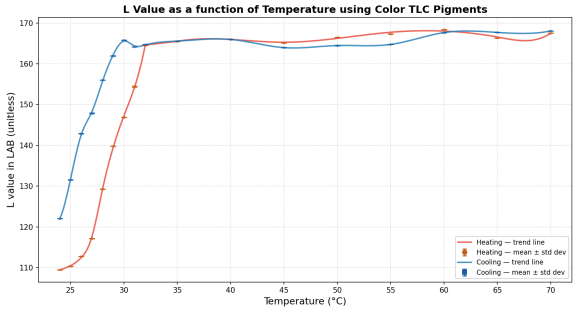
4.2 Temperature: Black Stripes

4.2.1 Experiment Setup

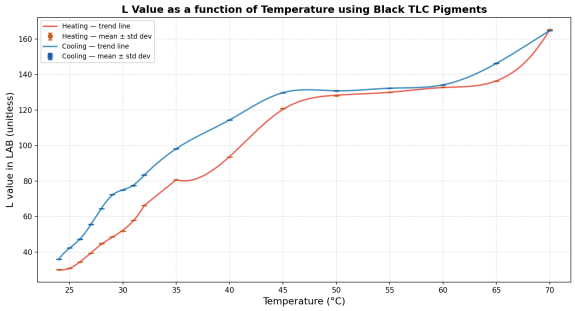
Similarly to the experiment done in the color model, a sample was also used with the same properties where a black layer of Psycho Paint (with 0.2mm) was glued to the acrylic glass. There was then the color layer with the same properties as detailed earlier and 0.5mm thickness which was superimposed on the black layer, this was to mimic the way the black grating is made which is 0.5mm of color layer on top of 0.2mm of the black layer. The number of samples taken at each temperature, as well as the range, steps, and dependent variable are exactly the same as previously.

4.2.2 Results

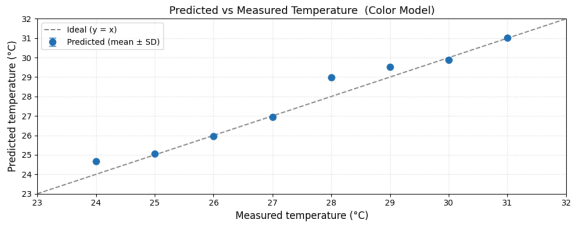
Figure 7b also shows a rise in the L value as the temperature increases throughout the entire range of temperatures, which is to be expected as the activation temperatures span that range. Once again, there is also a hysteresis loop between the heating and cooling runs.



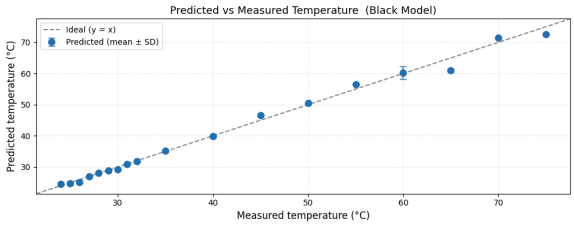
(a) The variation in the L value against temperature when using the color thermochromic pigments



(b) The variation in the L value against temperature when using the black thermochromic pigments



(c) The color model's predictions compared to the measured temperatures



(d) The black model's predictions compared to the measured temperatures

Figure 7

4.3 Temperature: Model

With the experiments done, which also show how the values of the pixels change with temperature, we can now exploit them to create a model that will allow us to predict the temperature of a point in the image by using an interactive window.

Firstly, a segmentation between the color and black stripes had to be done. This happens due to the colored stripes being much more sensitive within the narrower range defined in Section 2.4, but will contain no information above that temperature. On the other hand, the black stripes will be better able to predict temperature above that range and up until 70°C. The segmentation works essentially by looking at the bright spots and the dark spots row by row, and gets the median value for each purple and black stripe. It then assigns pixels to each one along each row based on its V value in the HSV color space.

Once the segmentation is done, the two models can then be made as when one clicks on the color stripe, it can then know to refer to the color model, and vice versa. The two models work exactly the same, with the difference being that they are more effective in different ranges.

The model works to estimate a temperature T based on the color of one of the two chosen stripes in the ROI. It does so as a non-parametric, instance-based regression where it interpolates among a labeled calibration set using inverse-weighted-distance k-nearest-neighbors. The particularity about this method is that it considers all three color spaces (RGB, HSV, CIELAB) with an automatic selection process based on choosing the color space with the smallest nearest neighbor distance.

In order for the automatic selection process to be unbiased, each channel c is min-max normalized following the formula:

$$\tilde{c} = \frac{c - m_c}{r_c} \tag{4}$$

where $\tilde{c}$ is the normalized pixel value of channel c, c is the pixel value of channel c, m_c is the mean of the channel and r_c is the range of the channel (so $max(c) - min(c)$).

The query color is chosen by a sampling method around where the user clicks, with the mean RGB being taken within a circular region of interest that has a radius r and center (c_x, c_y) (the point clicked). Sampling is performed in RGB so that the calculated mean can be projected into the different color spaces.

Within each of the different color spaces, the calibration points are used in order to calculate the Euclidean distance to the query point in the normalized frame. Then the k smallest are retained, with indices $j = 1, ..., k$, and being ordered as $d_{(1)} < ... < d_{(k)}$. The distance is calculated through:

$$d_i = \|\tilde{\mathbf{c}}_\mathbf{i}^{(s)} - \tilde{\bar{\mathbf{x}}}^{(s)}\|_2 \tag{5}$$

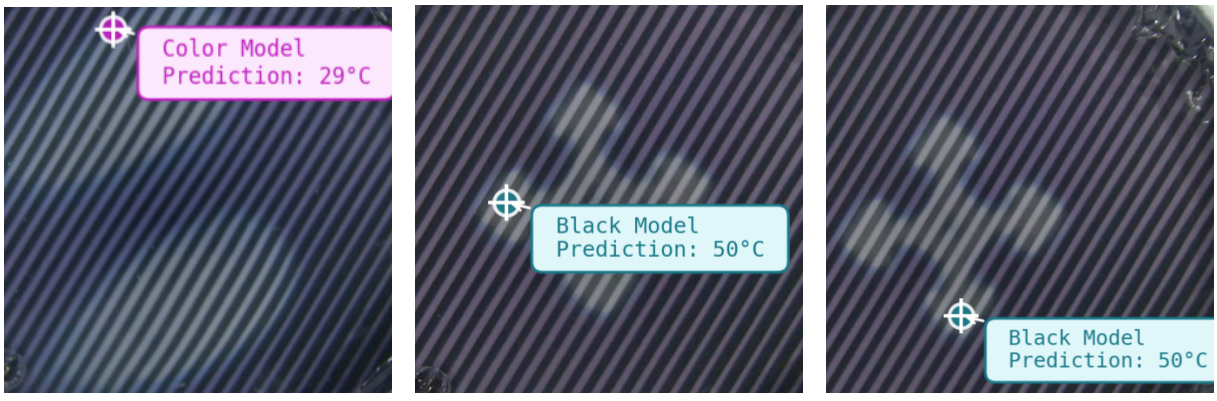
where d_i is the euclidean distance of calibration point i, $\tilde{\mathbf{c}}_\mathbf{i}^{(s)}$ is the normalized channel pixel values of calibration point i in color space s and $\tilde{\bar{\mathbf{x}}}^{(s)}$ is the normalized mean pixel values of the query point in color space s.

And the temperature is then estimated by using the inverse-distance-weighted mean of their labels, as shown below:

$$\hat{T} = \frac{\Sigma_j w_j T_{(j)}}{\Sigma_j w_j} \tag{6}$$

where $w_j = \frac{1}{d_{(j)}}$.

The result can then be seen in figure 8. Clicking on a color stripe will then inform you that it is therefore using the color prediction model and then estimates the temperature at the point of interest determined by the user.



(a) The color model being used to predict the outside of a human finger

(b) A metal piece heated to 50°C and predicted using the black model

(c) A different part of the heated metal piece showing consistent predictions

Figure 8: The different models being used to predict the temperature

Both the color and black model are then used to predict the actual temperatures in order to verify how well accurate each of them are over 10 images. Figure 7c has an R^2 of 0.9591 with a

RMSE of 0.46°C, while Figure 7d has an R^2 of 0.9958 and a RMSE of 1.07°C. This shows that both models are often able to correctly predict the right temperature.

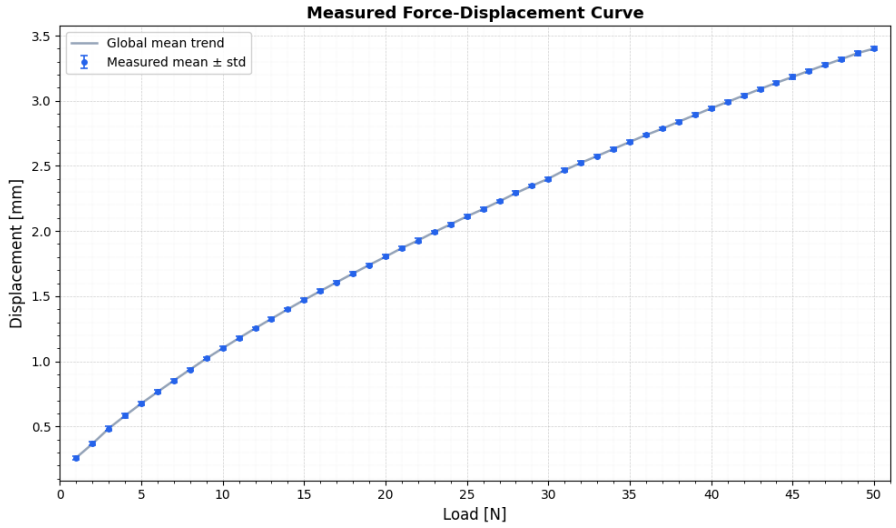
4.4 Force-Depth Relationship

4.4.1 Experiment Setup

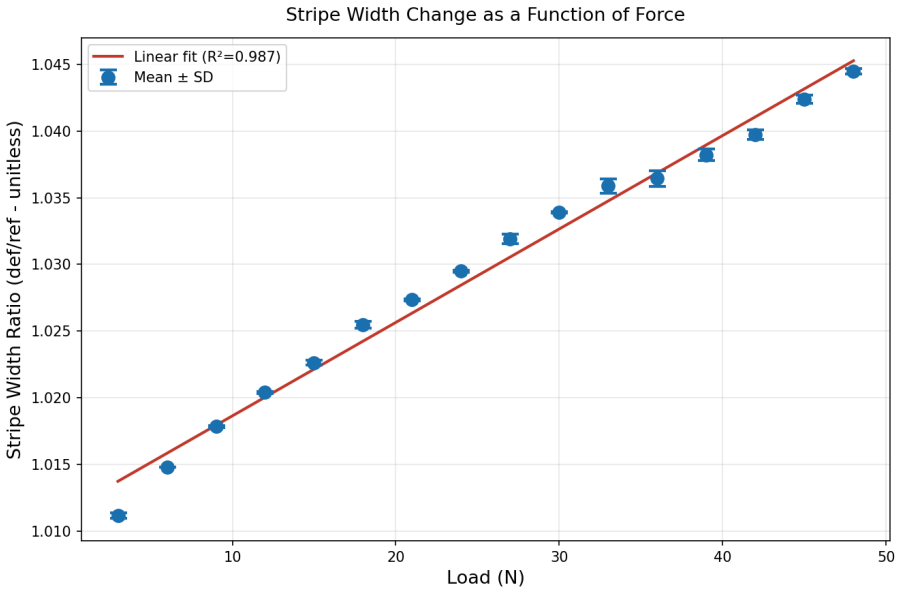
Using the sensor seen above and described as in Section 2, force was applied going from 0N to 50N (deemed to be the safe limit of the sensor) and the depth was measured continuously throughout by the Mark-10 Test Stand F-105 EM. This was run five times and values were taken at a step of 1N. For the experiment, a flat circular 3D printed piece (as can be seen in Figure 10a) was used to indent the sensor due to its simple shape and in order to keep the contact area constant.

4.4.2 Results

Force and displacement show a clear relationship, as can be seen in Figure 9a, where as one increases, so does the other. This explains itself due to the fact that as more force is applied, it would be expected that the indent would be pressed down further in the elastomer which is relatively soft. However, it is not a linear relationship and the gradient decreases as the load increases. This can be understood by knowing that elastomers are not purely elastic and where the more it is pressed down, the more the indent needs to displace an increasing volume of the elastomer.



(a) Force-Displacement Graph



(b) Force-Stripe Width Ratio Graph

Figure 9: Graphs showing the relationship between force and displacement, and then force and stripe-width ratio

4.5 Stripe Width Ratio-Force relationship

4.5.1 Experiment Setup

Using the same machine and 3D printed piece as in Section 4.4, a load was applied and then an image of the deformation was taken to then analyze. Forces ranging from 3N to 48N with a step of 3N were applied in order to determine how the width of the stripes change compared to when no deformation is applied. Three runs were performed for this experiment resulting in three deformed images per given load.

4.5.2 Analysis & Results

In order to be able to get results, the three deformed images were compared with a reference image where the area of contact was. The ratio between the width of the stripes when deformed compared to when they were not was taken and plotted against each of the respective loads. In order to look at how the stripe width compares, the carrier frequency was obtained for each of the image's ROI to then be able to obtain how many full fringe cycles could fit in that region. From there, the number of pixels across the ROI is obtained and divided by the number of full fringe cycles to get the period and the two periods are then used to calculate the ratio. Choosing to graph this with a ratio rather than the pure number of pixels per stripe was to illustrate how the shift isn't huge and therefore how noise could potentially affect the results.

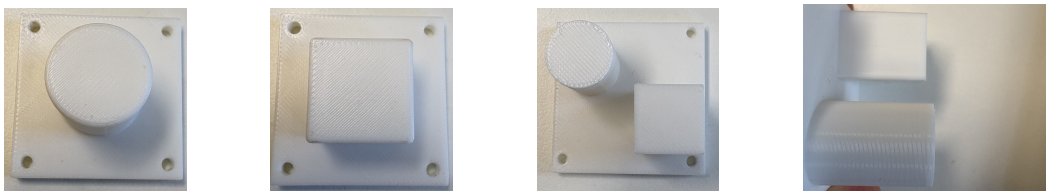
The results for that can then be seen in Figure 9b, with a clear trend that shows how the stripe width increases (and therefore the ratio increases) as a larger load is applied. This is further justified by the linear line of best fit giving an R^2 of 0.987 which shows a high degree of correlation. This is consistent with the fact that a larger force results in a higher displacement, which would stretch the elastomer and grating giving a change in the width of the stripes.

4.6 3D Reconstruction

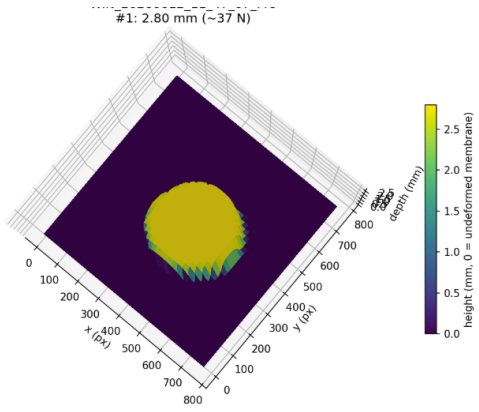
Using the results and data obtained from Sections 4.4 and 4.5, as well as the method to obtain shape in Section 3.1, it is possible to then reconstruct the deformation in three dimensions. Being able to detect the shape allows first a recognition of the deformation in 3D, and then through knowing how much the stripe width has then changed, the force can be obtained which then leads to the depth being known as well.

The algorithm for the 3D reconstruction relies heavily on the parts mentioned earlier, where first a shape detection is performed on the given image. Once the shape has been detected, the stripe width ratio is obtained for that area specifically (where the size for that is a slightly more restrained version in order to account for noise). From there a lookup table is used to then determine the correct depth for the shape detected (where the difference in area between the current shape and the base shape that was used for the experiments is taken into account), and the 3D deformation is then reconstructed as can be seen in Figure 10.

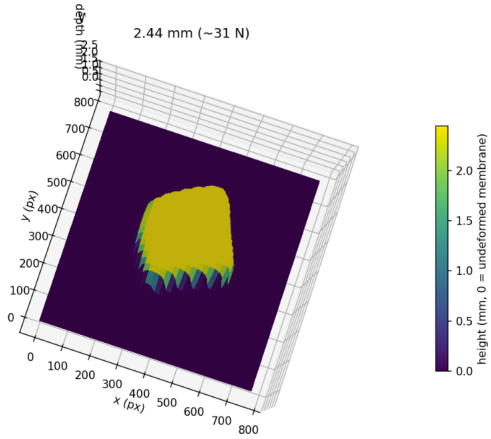
It is also possible to reconstruct different shapes within a single image as mentioned earlier, which then can lead to the lookup process happening for the two separate cases. This allows for shapes that have different depths to be accounted for and the visualization helps to show that difference.



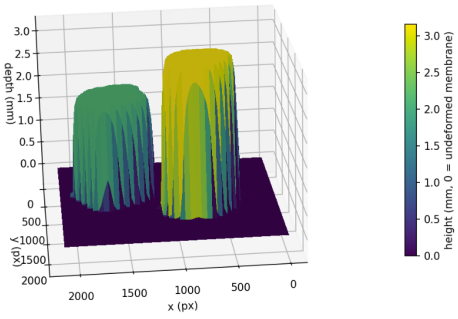
(a) The pieces for the deformation, where the last image shows that there is a difference in height between the circle and square



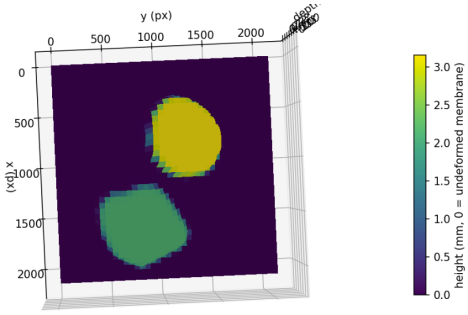
(b) Shape and Depth of a Circle



(c) Shape and Depth of a Square



(d) Side view of circle and square with different depths



(e) Top view of circle and square with different depths

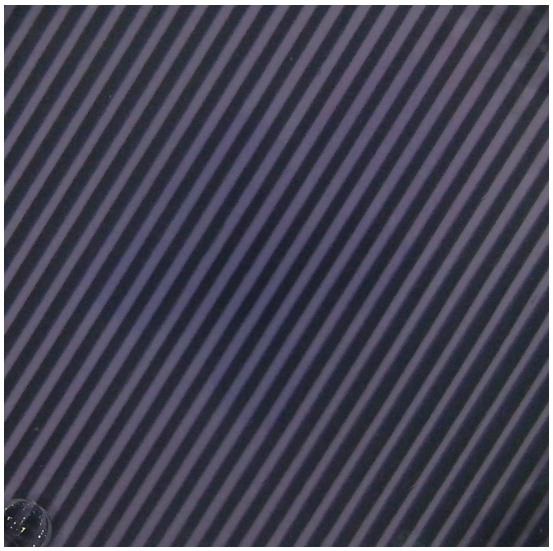
Figure 10: 3D reconstruction of two different shapes (square and circle) where it is possible to capture two separate depths (and shapes) in the same image

4.7 Integrated Pipeline

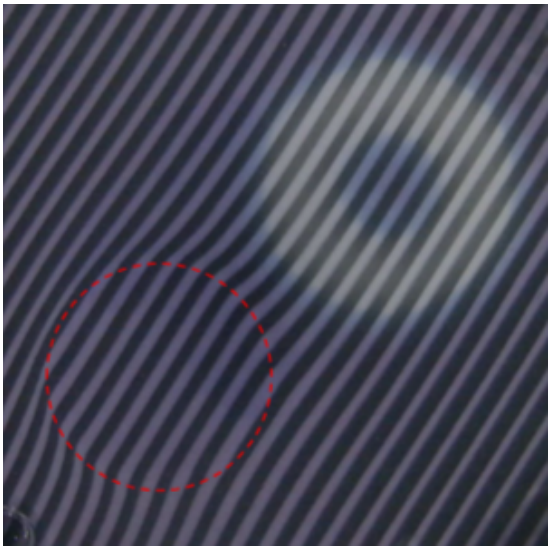
The final objective of this sensor is to be able to merge all the different aspects in a single pipeline, where one image with a heated object, and physical deformation by another object can all be detected when compared to a reference image. This shows that it is truly a multimodal sensor that requires only two images to be able to detect temperature, segment an object based on heat transfer, to recognize the shape, force and depth due to a physical deformation and to be able to reconstruct it in three dimensions.

This is clearly demonstrated by the results in Figure 12. All the different components are shown to work based on a single deformed image shown in Figure 11b which has a ring heated to 50°C placed on the sensor, and a circle that has been pressed into it (highlighted by a red dotted

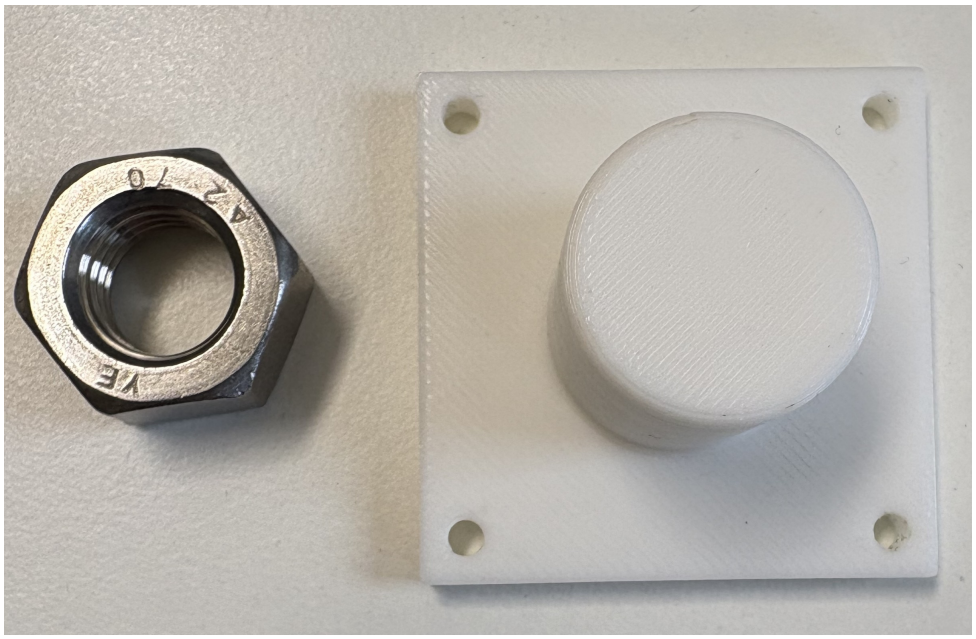
ring going around it. Figure 12b also demonstrates that the segmentation can detect a ring shape with the inner circle separating the unheated part in the middle from the rest. Using the black stripes enables the algorithm to correctly predict the temperature as shown in Figure 12a. Figures 12c and 12d then show the circle that was pressed in manually which is detected and then the depth is reconstructed with the force also estimated.



(a) Reference Image

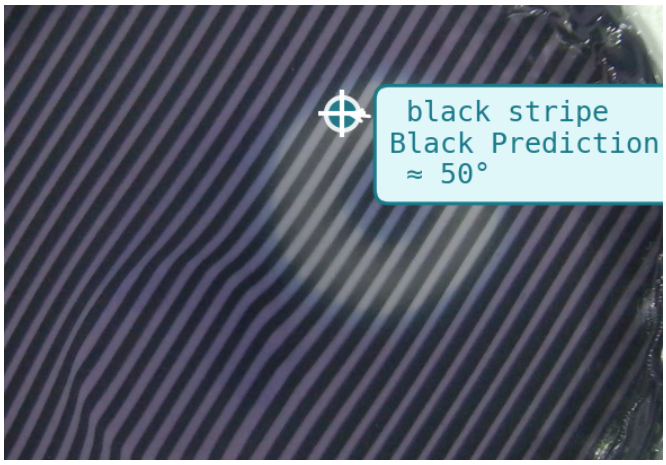


(b) Deformed Image where the dotted red circle was added for visual clarity

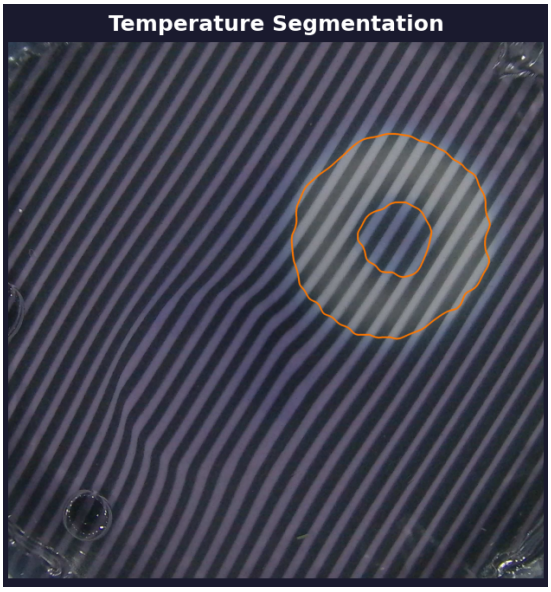


(c) The steel nut was heated while the 3D printed circle was used to cause the physical deformation

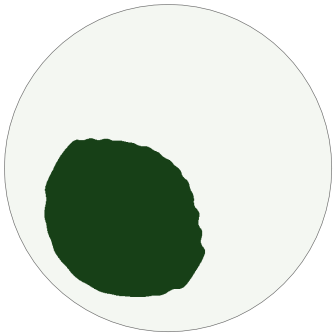
Figure 11: The two top images are what are put into the pipeline (without the red circle in the deformed image) in order to obtain the result. The bottom image shows what is what was used to obtain the deformed image



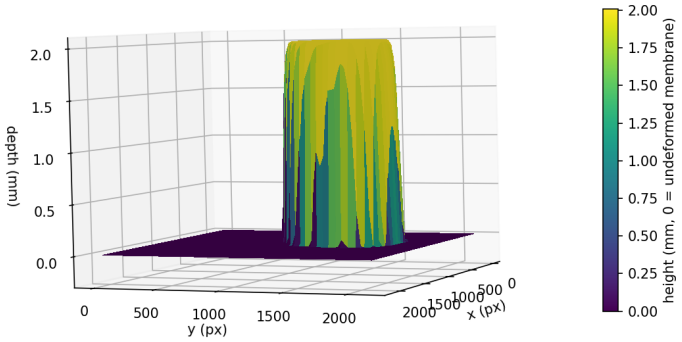
(a) Prediction of the temperature of heated ring



(b) Segmentation of the heated ring



(c) Shape detected due to physical deformation



(d) 3D reconstruction of circle with estimated depth and force

Figure 12: Overall pipeline that includes all previous methods that can be used to detect tactile and temperature information based on a single image

5 Potential Applications

Overall the sensor shows great flexibility and an ability to quite accurately detect various temperature and tactile components. One promising application would be to use it on a robotic hand, which could lead to more precise handling of objects, as mentioned earlier in Section 1. In this context, it could be very useful on the tactile side, with a list of potential differentiations that it could make based on the 3D reconstruction, shown in Section 4.6. These are shown in Table 2. For instance, the hand could be used to detect the same object, but with different forces applied. It could also detect the same shape, but in different sizes, and finally it could detect differences in height of otherwise similar objects.

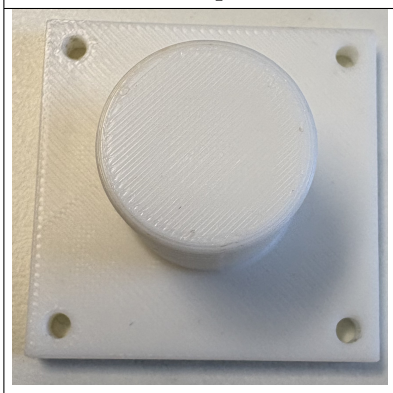
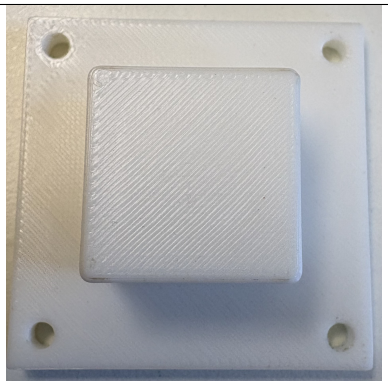
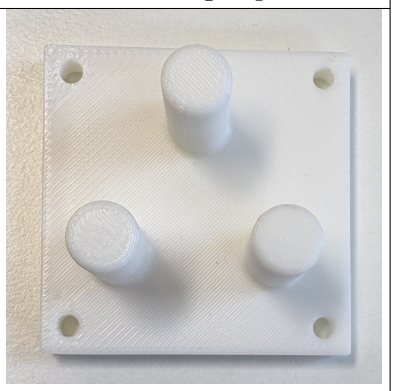
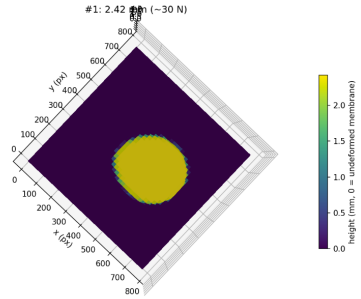
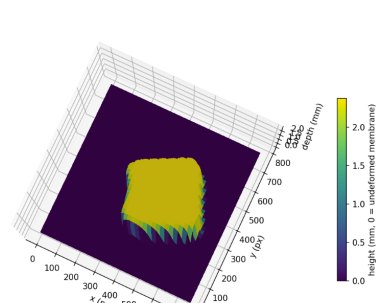
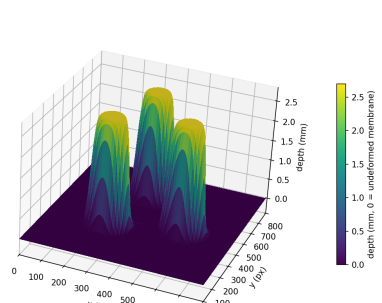
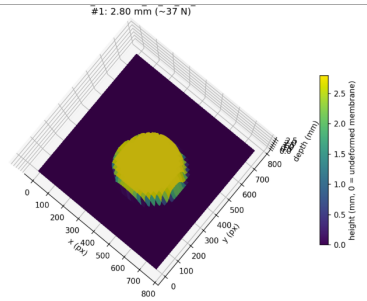
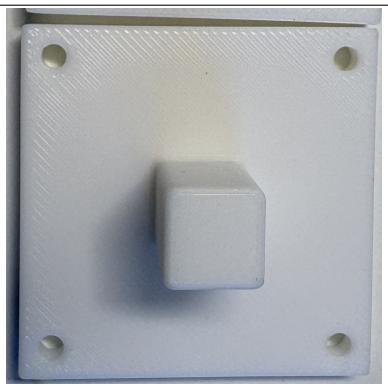
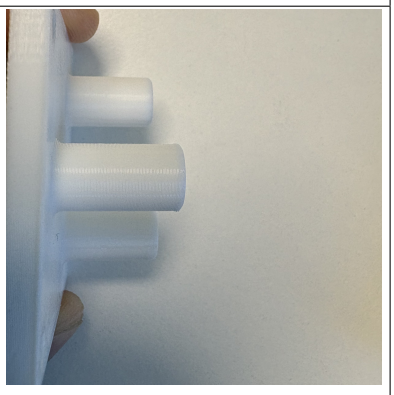
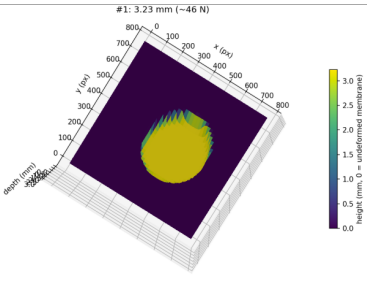
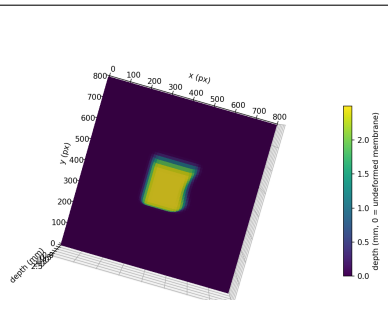
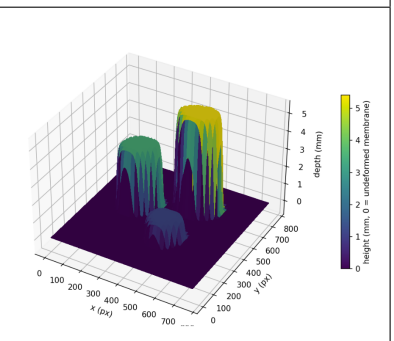
Same shape, different forces and depths	Same shape, different sizes	Same configuration, alternating depths
		
 Depth=2.42mm & Force=30N		
 Depth=2.80mm & Force=37N		
 Depth=3.23mm & Force=46N		

Table 2: Demonstration of different potential applications of the sensor, for example while on a robotic hand

6 Conclusion

This project presented a physical decoupled programmable skin, that can use just two photos (one reference and one with deformations) from a single camera to detect temperature, segment heated objects, detect the shape, force and depth of physical deformations and be able to reconstruct those in a 3D visual.

The thermochromic pigments were able to show clear changes in color at the different temperatures and a trend between the L value and temperature was demonstrated, allowing a model to be built that could predict the temperature with both high sensitivity and high range. This shows the potential for interactions with humans and with objects at high heat. It could also be used to have a better understanding of an object's material that it is coming into contact with, due to its thermal capacity.

Using FTP also led to being able to detect shapes of different kinds, and to detect several shapes in a single deformed image. A link between force, depth and the change in the width of the stripes was established, which allowed us to build on that and use it to give a visualization in three dimensions. This shows the potential of the sensor in being able to go further and eventually reconstruct more challenging shapes and 3D figures such as a mountainous landscape for example.

There are certain improvements that could be made to push the sensor's capabilities, especially with regards to upgrading the camera's resolution and spending more time trying to develop the manufacturing to make the grating thinner and the period of the stripes lower as well. A smaller period for the stripes would most likely lead to a more precise FTP and therefore a better shape detection and 3D reconstruction. A higher resolution camera would also lead to a better visualization of the edges of the stripes which would also improve the tactile side.

7 References

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I would also like to thank the CREATE Lab and Professor Josie Hughes for allowing me to do my thesis here and for providing so many tools at our disposal that allows for the success of many projects.

A Appendix

A.1 Brightness of the LED strip

Different PWM values were tested to see which one would allow for the greatest contrast by using the Michelson Contrast. The results can be in Figure 13. The PWM was increased in steps of 15, the graph shows that a lower value gives a higher Michelson contrast and therefore a PWM of 15 was chosen.

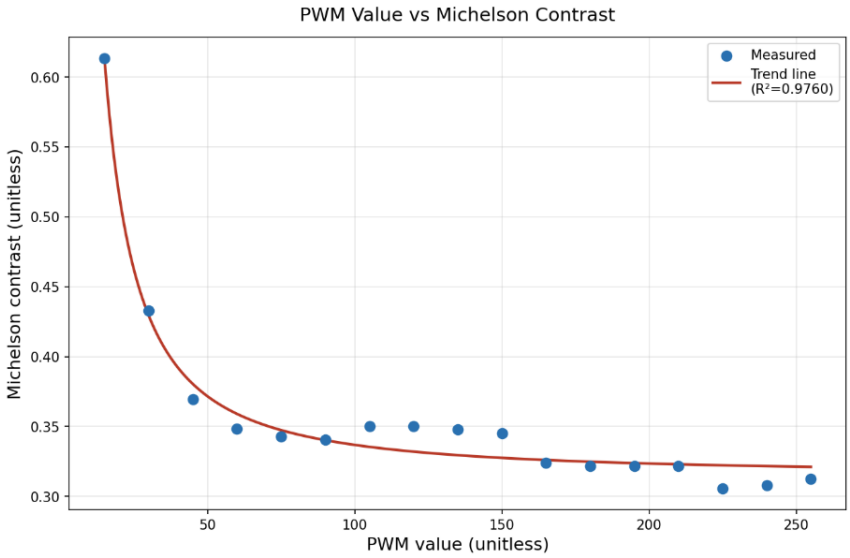
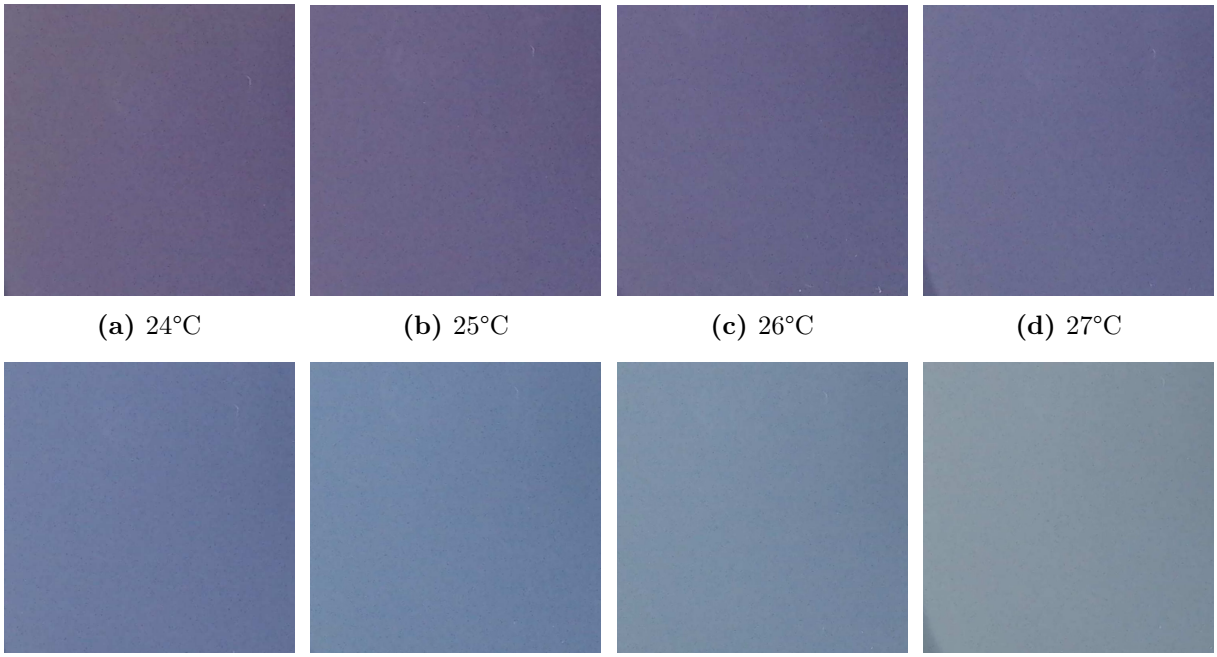


Figure 13: Graph that shows how the Michelson Contrast decreased as the PWM value increased and then saturated. Having a smaller PWM helped to make the difference between the stripes in the grating cleaner

A.2 Colors observed in the Temperature Experiments

All the colors in the sensitive range (24°C-31°C) using the colored thermochromic pigments are shown below in Figure 14.

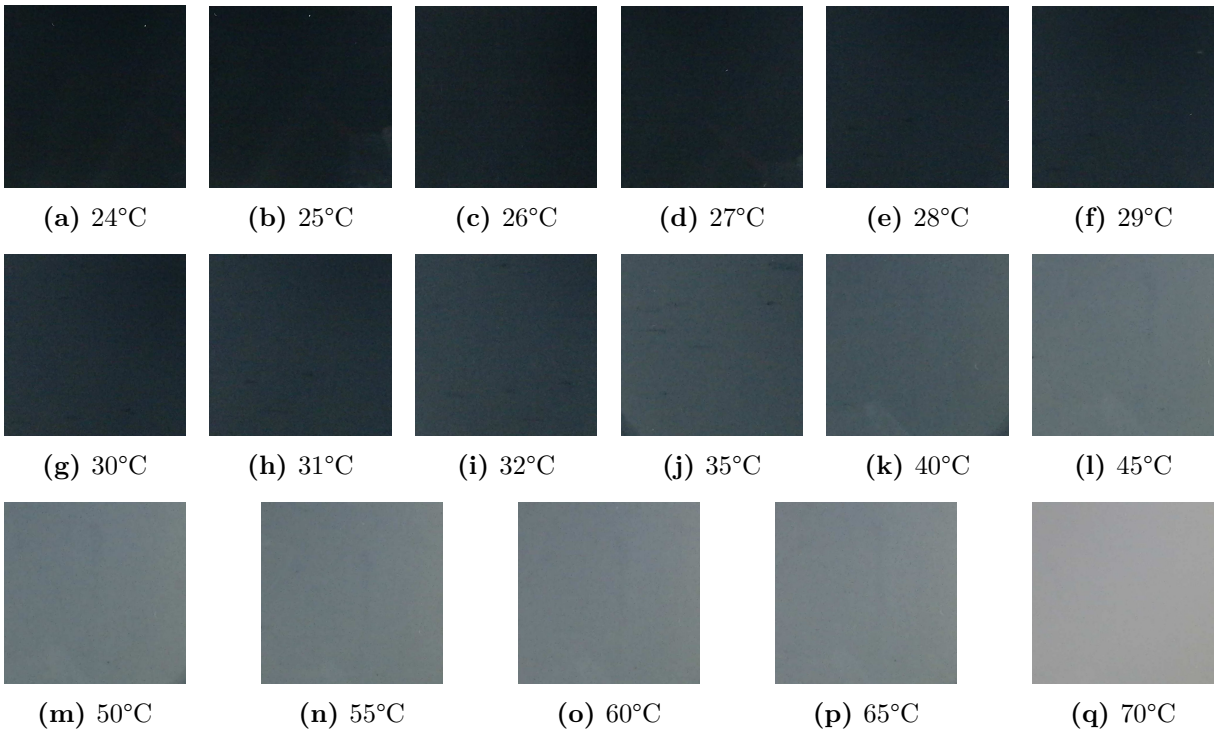


(a) 24°C **(b)** 25°C **(c)** 26°C **(d)** 27°C

(e) 28°C **(f)** 29°C **(g)** 30°C **(h)** 31°C

Figure 14: The different colors observed in the sensitive range (24°C-31°C) for the colored thermochromic pigments

All the colors taken when doing the experiments with the black thermochromic pigments are shown below in Figure 15.



(a) 24°C **(b)** 25°C **(c)** 26°C **(d)** 27°C **(e)** 28°C **(f)** 29°C

(g) 30°C **(h)** 31°C **(i)** 32°C **(j)** 35°C **(k)** 40°C **(l)** 45°C

(m) 50°C **(n)** 55°C **(o)** 60°C **(p)** 65°C **(q)** 70°C

Figure 15: The different shades of black observed in the wide range (24°C-70°C) due to black thermochromic pigments

A.3 Masses used for fabrication

For the elastomer, 1.5g of curing agent was used with 45g of PDMS. For the DragonSkin20, 0.5g of yellow pigments, 0.3g of red pigments, and 0.3g of blue pigments was used to make 20g of silicone. For the Psycho Paint, 0.12g was used for the black pigments with activation temperatures of 10°C, 18°C, 38°C, 45°C, and 70°C, and 0.15g for the rest.

A.4 Code

The code and other relevant files can be found at the following GitHub link: https://github.com/liambgibbons/master_thesis